

Sulagna Ghosh

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Employment History

- January 2023 – Present  **Teaching assistant**, The University of Chicago. Part of Ph.D. funding; assisted in both undergraduate and graduate courses in Statistics, holding weekly office hours, leading problem sessions, and grading assignments and exams.
- July 2020 – August 2022  **Risk Modeling and Analytics Specialist**, UBS Business Solutions (India) Pvt. Ltd., Mumbai, India. Worked for the AI, Monitoring & Surveillance (AIMS) team, validating financial models aimed at monitoring transactions and/or detecting suspicious activities.
- June 2019 – August 2019  **Project trainee**, Centre for infrastructure, Sustainable Transportation and Urban Planning (CiSTUP), Indian Institute of Science, Bengaluru, India. Worked on choice models with stochastic variables and random coefficients used for the analysis of transportation data.

Education

- September 2022 – June 2027  **Ph.D. in Statistics, The University of Chicago, Chicago.**
Research area: *Applying machine learning tools with Bayesian inference methods in different real-life datasets.*
- July 2018 – June 2020  **M.Stat., Indian Statistical Institute, Kolkata.**
Specialization in Applied Statistics (Finance). Overall Percentage: 91.2%.
Thesis title: *Study on retrospective inference from recalled time-to-event data.*
- August 2015 – June 2018  **B.Sc. in Statistics, Presidency University, Kolkata.**
Cumulative GPA: 9.81 out of 10.

Current Projects

-  **Flexible and scalable Bayesian modeling of spatiotemporal event data with low-rank Neyman-Scott tensor processes**
Supervisor: Aaron Schein (Assistant Professor, Department of Statistics, The University of Chicago, Chicago)
Consider data of the form “news source s reports that actor i took action a toward actor j at time t ”. To fit a temporal point process model to such large-scale event data, a gradient-based scalable model is proposed to infer the latent events that are interpretable as “major stories” which generate related events over long periods. While tailored to international relations data, our work is broadly applicable to marked temporal event data, and generally builds upon recent advances in neural point processes, probabilistic non-negative tensor decomposition, and scalable approximate Bayesian inference.
-  **Study on retrospective inference from recalled time-to-event data**
Supervisor: Debasis Sengupta (Professor, Applied Statistics Unit (ASU), Indian Statistical Institute, Kolkata)
A careful modeling approach is proposed, accounting for both inaccuracy and imprecision in recall, addressing the issue of confounding between a correctly and incorrectly recalled data. In case there is very low amount of recall error, the model works competitively with the model considering only recall imprecision, while the model performs much better than the existing model when there is significant amount of recall error present. Applied to a real-life dataset collected on selected females who were asked to recall the date of onset of their menarche.

Research Articles

- 1 S. Ghosh and A. Schein, *Scalable amortized variational inference for non-poisson buy-'til-you-die models*, <https://arxiv.org/abs/2608.19022>, **Under review at IJRM**, 2026.
- 2 S. Ghosh, N. Ignatiadis, F. Koehler, and A. Lee, *Stein's unbiased risk estimate and hyvärinen's score matching*, <https://arxiv.org/abs/2502.20123>, **Under revision at JRSS-B**, 2025.
- 3 M. Biswas, C. R. Bhat, S. Ghosh, and A. R. Pinjari, "Choice models with stochastic variables and random coefficients," *Journal of Choice Modelling*, vol. 51, 2024.  URL: <https://doi.org/10.1016/j.jocm.2024.100488>.

Skills

Programming Languages  Python, R, C, Gauss, MATLAB, $\LaTeX$, Microsoft Excel, Git.
Languages  Full professional proficiency in English, Bengali, Hindi.

Miscellaneous Experience

Awards and Achievements

- 2026  Received a **Best Poster Prize** at the conference **ISBA 2026 World Meeting** in Nagoya, Japan, awarded for the poster titled "Flexible and scalable Bayesian modeling of spatiotemporal event data with low-rank Neyman-Scott tensor processes".
- 2020  Shortlisted for the competition for **P. C. Mahalanobis Memorial Gold Medal**, which is awarded for outstanding performance in M.Stat. by Indian Statistical Institute, Kolkata.
- 2018-2020  **Indian Statistical Institute Teachers' Committee Awards for Outstanding Academic Performance**, in both years of M.Stat. at Indian Statistical Institute, Kolkata.
 Received **full Stipend/Fellowship** from Indian Statistical Institute throughout M.Stat.
- 2018  Secured **All India Rank of 1** in M.Stat. entrance examination of Indian Statistical Institute.
 Secured **4th position** in All India IIT-JAM Examination for Mathematical Statistics.
 **Dr. Syama Prasad Mookherjee Gold Medal for Scholastic Record in Undergraduate 2018**, Presidency University, Kolkata.
 **Presidency Alumni Association Gold Medal for highest marks in the Bachelor of Science 2018**, Presidency University, Kolkata.
 **Gold Medal for securing first position in Bachelor of Science (Statistics)**, Presidency University, Kolkata.
- 2015-2018  **INSPIRE-SHE Scholarship for being among the top 1% in the 12th board examinations** issued by the Department of Science and Technology (DST), Government of India.
- 2004  **Award of Excellence in the Junior Diploma Examination of Drawing** from Calcutta Talent Search School, Kolkata.

Extra-Curricular Activities

- 2025-2026  **Co-president of Kheyal**, the Bengali student organization at The University of Chicago - Deciding and organizing social events.
- 2024-2025  **Decoration lead of Kheyal**, the Bengali student organization at The University of Chicago - Led decorations for several events organized by Kheyal.
- 2021-2022  **Mentor for new joiners at UBS Business Solutions (India) Pvt. Ltd.**, Mumbai - Helped new team members with their training during the training week and guided and helped them settle into their role.